



SMART-SECURE WATER

LEAKAGE DETECTION & MANAGEMENT

“TEAM OBLIQUES”

MANISH, MITHIL, JAWAD, JASPRATAP

Project Overview

Modernizing water utility management through an integrated sensing ecosystem.

Our system provides a holistic approach to resource security by combining automated leakage detection, tank level monitoring, and Bluetooth-based access control. It transforms a passive resource into an active, intelligent asset that prevents waste and ensures authorized consumption only.



| System Objectives



Leakage Detection

Immediate identification of pipeline breaches using high-precision IR sensors to trigger rapid alerts.



Resource Tracking

Monitoring tank levels and flow rates in real-time to provide precise data on utility consumption.

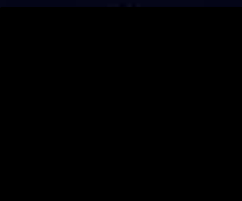


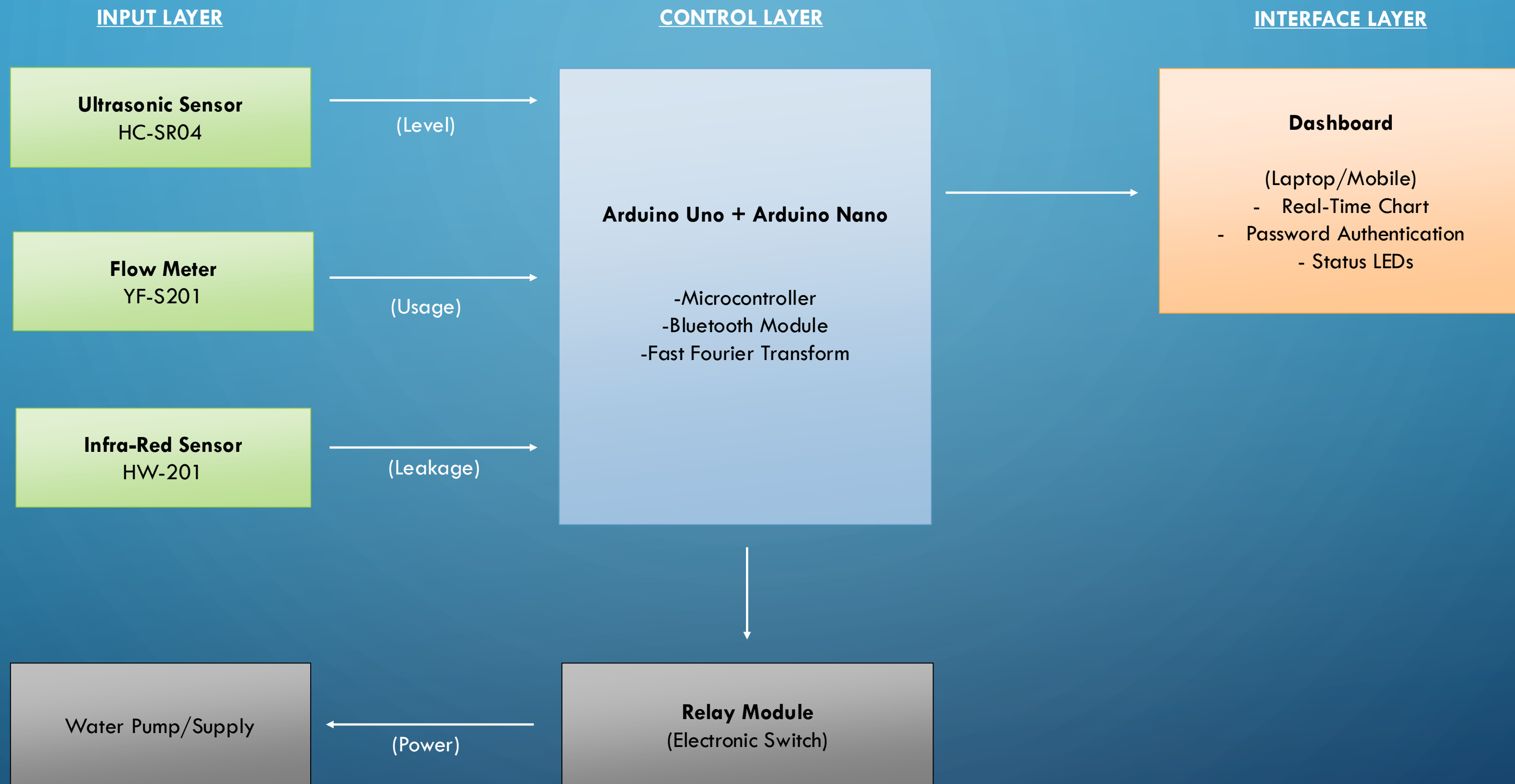
Access Control

Implementing a password-protected Bluetooth authentication layer to control solenoid valves.

Hardware Components

Category	Component	Primary Function
Central Processor	Arduino UNO / Nano	Core logic processing and sensor data aggregation.
Environment Sensing	Ultrasonic & IR Sensors	Tank level measurement and physical leakage detection.
Fluid Monitoring	Water Flow Sensor	Quantifying real-time flow rate for abnormal pattern detection.
Connectivity	HC-05 Bluetooth Module	Wireless alert transmission and command reception.
Access Control	Relay & Solenoid Valve	Electronic actuation of water supply based on auth.





Operational Workflow

1. Continuous Sensing

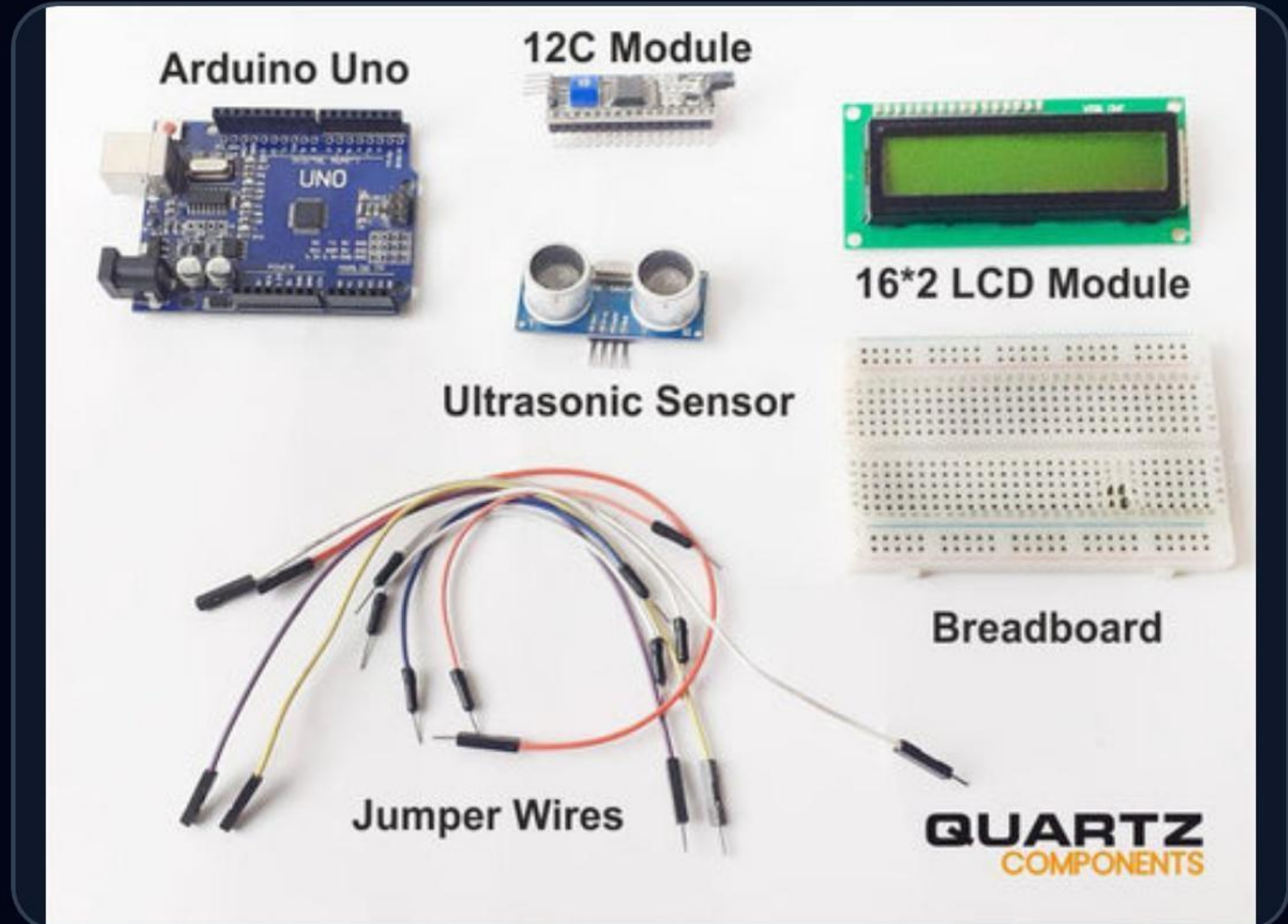
Sensors monitor the pipeline and tank environment 24/7, relaying data to the Arduino microcontroller.

2. Anomaly Trigger

If ultrasonic values drop or flow exceeds thresholds, a Bluetooth alert is instantly broadcasted.

3. Secure Actuation

Valve remains locked. Access is granted only upon receipt of the correct Bluetooth password string.



| Uncompromising Security

100%

Secure Access

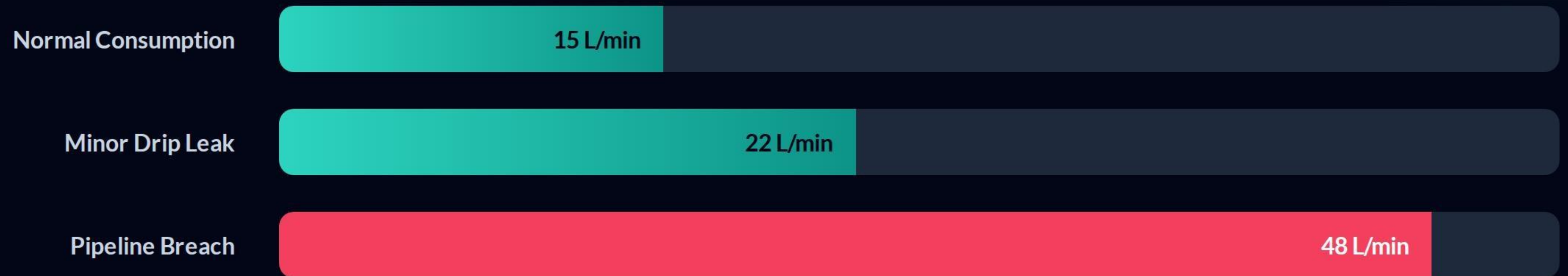
Protection Against Unauthorized Use

The system implements a robust password-based authentication protocol via Bluetooth. Without the correct credentials, the solenoid valves remain closed, effectively preventing water theft and tampering in shared or industrial environments.

| System Capabilities

-  **Real-Time Response:** Instantaneous detection and notification of pipeline leaks or abnormal flow spikes.
-  **Wireless Interface:** Seamless monitoring and control via any Bluetooth-enabled terminal or mobile application.
-  **Automated Fail-Safes:** Capability to automatically shut off valves if a critical leakage pattern is identified to prevent flood damage.
-  **Consumption Insights:** Precision tracking of daily usage volume for improved budgeting and resource planning.

Flow Pattern Analysis



Comparative flow rate thresholds utilized by the system logic to differentiate between usage and failure.

Future Scope: Smart Economics

Consumption Forecasting

Utilizing machine learning to analyze historical data and predict future demand. This allows for optimized tank filling during off-peak electricity hours, reducing operational costs.

Automated Billing Systems

Creation of "Water Wallets" for multi-resident complexes. Integrated billing triggers automated supply cutoff when pre-paid balances are exhausted, ensuring fair utility usage.

💡 Aiming for a fully autonomous and self-sustaining water economy.

| Future Scope: Quality & Acoustics



Safety Beyond Quantity

We plan to integrate pH, Turbidity, and TDS sensors to ensure the water is safe for consumption. The system will alert users if contaminants are detected in the tank.

Acoustic Leak Pinpointing

Using specialized sensitive microphones to detect the unique acoustic signature (hissing) of underground leaks that physical flow sensors might miss.

The background is a solid blue gradient. In the corners, there are decorative white line art elements resembling circuit boards or neural networks, with lines and small circles connecting them.

"This system provides an intelligent and secure water management ecosystem, bridging the gap between basic utility monitoring and advanced digital security."

~System Design Philosophy